

Calculus – Group Discussion

Week 1 – Limits, Continuity, and the Derivative

ADSAI & CS / PR12

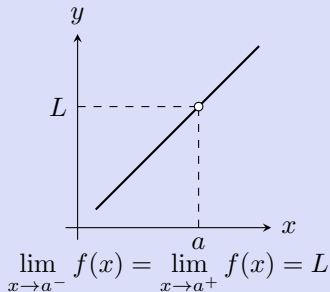
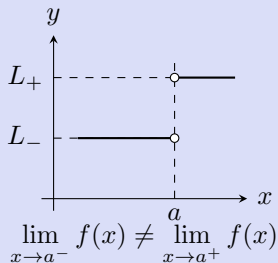
Learning goals (Week 1)

- Interpret limits graphically, numerically, and algebraically.
- Apply limit laws and algebraic simplification to compute limits.
- Decide whether a function is continuous at a point.
- Understand the derivative as the limit of a difference quotient.
- Differentiate polynomials using first principles and the power rule.

Limits

A function $f(x)$ has limit L at $x = a$ if and only if it has both left and right sided limits there and these one-sided limits are both equal to L :

$$\lim_{x \rightarrow a} f(x) = L \iff \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = L$$



Visualization: <https://www.desmos.com/calculator/ba7uaf4tey>

Continuity

Continuity at an interior point

A function $f(x)$ is **continuous** at an interior point c of its domain if

$$\lim_{x \rightarrow c} f(x) = f(c)$$

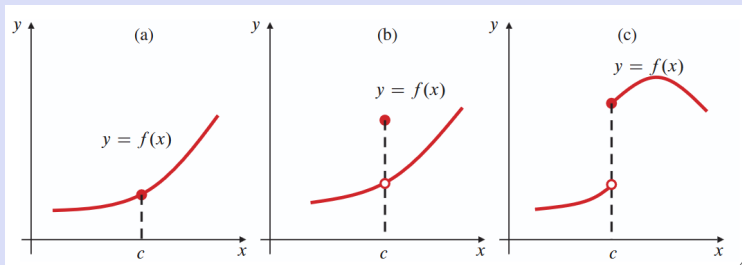
f is **discontinuous** if

- (1) $\lim_{x \rightarrow c} f(x)$ fails to exist, or;
- (2) it exists but is not equal to $f(c)$.

(a) f is continuous at c

(b) $\lim_{x \rightarrow c} f(x) \neq f(c)$

(c) $\lim_{x \rightarrow c} f(x)$ does not exist



Derivative as the limit of a difference quotient

The **derivative** of a function f is another function f' defined by

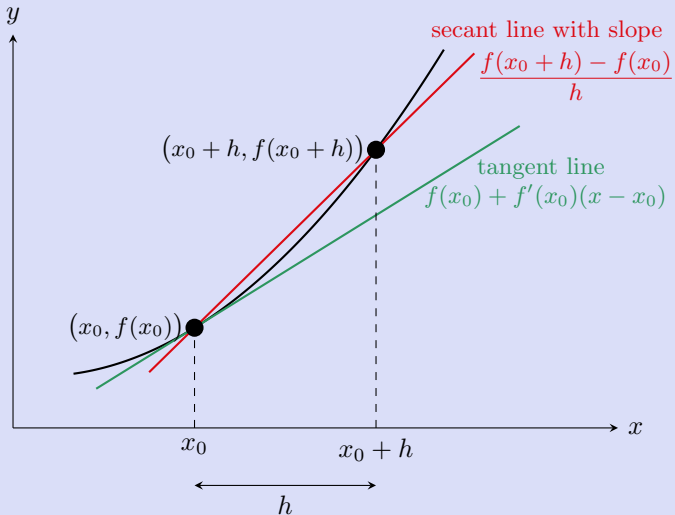
$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

at all points x where this limit exists.

Interpretation of $f'(x)$: how much does the output $f(x)$ change when the input x changes by a small amount?

Other derivative notations:

$$f'(x) = \frac{df}{dx} = \frac{d}{dx}f(x) = y' = \frac{dy}{dx}$$



Key takeaways

- A limit describes the behaviour of a function near a point.
- Algebraic simplification often reveals the limit hidden behind an undefined expression.
- Continuity at $x = a$ requires both the limit to exist and the value to match it.
- The derivative comes from the limit of a difference quotient.
- The power rule is a practical consequence of the derivative definition.